



Shotha

Learning Objectives

- ⇒ To learn the concept of Shotha
- ⇒ To know the Nidana of Shotha with current understanding
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To enlist the types of Shotha
- ⇒ To describe the Vidhi Samprapti of Shotha
- ⇒ To describe Purvarupa, Vishishta Lakshana, Sadhyasadyata of Shotha with current understanding
- ⇒ To differentiate Doshaja Shotha

Introduction

Shotha is the spectrum of the diseases which is having the cardinal feature of *Utsedha* (swelling). This is considered to be manifested due to the involvement of *Rasa Dhatu*. This can happen either locally or half of the body or generalized.

Depending upon the site of the lesion, we can able to predict the involvement of different *Dhatu*. As far as affliction of *Srotas* is concerned, *Rasavaha Srotas* is primarily afflicted with *Sanga* and *Vimargagamana* as the nature of *Srotodushiti* happening during the course of the disease.

सर्वमुत्सेधं शोफमित्याहुः

Atanka Darpana, Ma.Ni. – 36/1,2

Any kind of swelling occurring in any part of the body is

termed as *Shopha* or *Shotha*. It may be either localised or generalized.

Nidana of Shotha

शुद्धयामयाभुतकृशावलानां क्षाराम्पतीरणोष्णगुरुपसेवा।

दध्यामसुच्छाकविरोधिदुष्टगरोपसृष्टान्न निषेवणं च॥४॥

अशस्यचेष्टा न च देशुन्दिर्ममषघातो विषमा प्रसूतिः।

मिथ्योपचारः प्रतिकर्मणां च निजस्य हेतुः श्वयोः प्रदिष्टः॥५॥

Ma.Ni. – Shotha/4-5; Ca.Cl. – 12/5-6

- ⊗ In case of administration of **purificatory treatment**¹
- ⊗ As a secondary factor of **other illness**²
- ⊗ When the person is doing fasting or during **starvation**³

- 1 The medications which are used for purificatory therapy (Shodhana) will have Ushna, Tikshna, Sukshma, Vyavayi and Vikasi qualities. These are responsible for the inflammation of the alimentary mucosa. The medicines used for Vamana (emesis) will lead to the inflammation of upper alimentary canal. The medicines used for Virechana (purgation) will lead to the inflammation of lower alimentary canal, small intestine in particular. The medicines used for Basti line of treatment, will lead to the inflammation of colon or large intestine.
- 2 Whenever an individual is suffering from any type of either acute or chronic illness, that will activate the leucocytes which in turn will produce interleukins. These interleukins are proteins made inside the body due to the exposure of the body to the pathogens. Let us take for example, if an individual is having the trauma, that will stimulate the leucocytes of the body to produce excess interleukins in the interstitial compartment and this will increase the oncotic pressure of the interstitial compartment and this will lead to the excess accumulation of fluid in the interstitial compartment leading to the formation of oedema.
- 3 Even though the tissues of the body preferentially use carbohydrates for energy over both fat and protein, the quantity of carbohydrate normally stored in the entire body is only a few 100 gms (mainly as glycogen in the liver and muscles), and it can supply the energy required for body functions for only half a day. Therefore, except for the first few hours of starvation, the major effects of metabolism are sequential depletion of fats and proteins. The final outcome could occur is protein depletion in the body even in the blood vessels too. Thus, in severe form of starvation, the person may develop slow pitting type of oedema.

- When the person is **emaciated**⁴
- When the person is having **weakness**⁵ or reduced physical strength
- When the person is consuming **Kshara**⁶ (alkaline) substances
- When the person is consuming excess **Amla** (sour)⁷ substances
- When the person is consuming excess **Tikshna** (irritating or chilli) like substances⁸
- Excess consumption of **Guru**⁹ (heavy) substances
- Excess consumption of **Dadhi**¹⁰ (curds)
- Excess exposure to **Ama**¹¹
- Consumption of **Mrit** (mud)¹²
- Excess consumption of **Shaka** (vegetables)
- Excess consumption of **Virodhi Ahara**¹³ (substances which are having opposite qualities)
- Consumption of **Dushta Ahara**¹⁴
- Consumption of food substances having **Gara Visha**¹⁵
- Patient of **Arshas**¹⁶
- Person having **sedentary life style**¹⁷
- When the person is not having **proper cleanliness** of

- 4 When an individual is presented with too much emaciation, there is a possibility of reduction in the proteins in the body. This implies there will be reduction in the proteins in the blood vessels too. This results in the reduced oncotic pressure in the capillaries. When there is reduction in oncotic pressure, that will lead to movement of liquid portion from the capillaries to interstitial space. Thus, an individual will have slow pitting type of oedema.
- 5 In this type of etiological factor, the mechanism will be similar to that of starvation.
- 6 The consumption of Kshara, Amla and Tikshna substances will lead to the increase of Tejo Mahabhuta in the body and thus this is responsible for the increase of Pitta in the body because of Ushna Virya.
- 7 Excess consumption of Amla Rasa substances will increase the hydrogen ions in the stomach leading to the further acidification of gastric juices. These juices are considered to be responsible for the erosion of gastric mucosa and thus will cause inflammation of gastric mucosa.
- 8 As these substances stay more time in the stomach for about two hours, these will cause inflammation of stomach and this will lead to vomiting. Due to the increased peristaltic movements of the stomach, these substances will move into the small intestine causing inflammation of small intestine and will lead to diarrhoea.
- 9 Excess consumption of Guru (heavy) substances will lead to the increase of Kapha Dosha in the body based on the concept of Samanya Vishesh Siddhanta. As the person consumes Guru substances, it may accumulate in the stomach for more than the required amount of time. This may lead to the infection followed by inflammation of the stomach. Swelling of the epithelial cells of the stomach may be observed as one of the signs of inflammation, the other signs of inflammation include pain, local raise of temperature, redness.
- 10 Dadhi is considered to be having Madhura and Amla Rasa. Madhura and Amla Rasa may lead to the vitiation of Kapha and Pitta Dosha. Amla Rasa may lead to the vitiation of Pitta Dosha. Amla Rasa substance may lead to the inflammation of gastric mucosa. One of the main features of inflammation is swelling, other features are pain, local raise of temperature, redness.
- 11 Refer the context of Ama mentioned in Hridroga for the description of Ama in the body.
- 12 The people who are severely undernourished can have aberration of diet including pica (consumption of non nutritive substances such as clay, charcoal, ice, dirt or paint). This is usually seen in patients who are having anemia. The person who are presented with anemia, may have deficiency of protein, minerals and this phenomenon leads to the reduction of oncotic pressure and this could be the reason for the development of oedema.
- 13 When an individual consumes more and more quantity of Viruddha Ahara (incompatible substances), this will lead to the inflammation of alimentary canal, lower alimentary canal in particular.
- 14 This can be attributable to severe vomiting and diarrhoea with prostration (act of lying down flat) can be observed with 2-4 hours ingestion caused by a toxin present in the food. This is observed in food poisoning. The foods contaminated are usually cooked meat or pies containing meat and starches. In this situation, we can observe the inflammatory reaction in the stomach and intestines. One of the features of inflammation is swelling.
- 15 गरसंयोगजं चान्यद्रसंज्ञं गरप्रदम्। कालान्तरविपाकित्वान्न तदाशु हरत्यसूनु॥
गरार्थः संयोगो येषां ते गरसंयोगा द्रव्यभेदाः, तेभ्यो जातं गरसंयोगजम्। तच्च चिरकारिरोगजनकप्रमाणहरम्॥
गरं तु द्विविधं - निर्विशद्रव्यसंयोगकृतं, तथा सविषद्रव्यसंयोगकृतं च, तत्राद्यं गरसंज्ञम्, उत्तरं तु कृत्रिममिति व्यवस्था।
Visha is any substance which produces discomfort to the patient. Gara Visha is the one which is considered to be non toxic, but consumption of Gara Visha is responsible for the adverse effects which may be produced after a long time and it will not cause death of the individual. Gara Visha is of two types - Savishadravyayukta (mixed with toxic substances) and Nirvishadravyayukta (not mixed with toxic substances). As these substances may stay in the body for longer time, that may lead to the inflammation of the alimentary canal in due course of time. As already stated, swelling is one of the features of inflammation.
- 16 The term Arshas refers to varicosity of haemorrhoidal veins. In the patient of Arshas, we can either see the swelling of the anal region which could be due to thrombosis of haemorrhoidal veins or the ascites (accumulation of fluid in the peritoneal cavity and this is attributable to portal hypertension).
- 17 To understand this concept, we need to learn the physiology of lymphatic system. Whenever, there is entry of excess amount of proteins in the interstitial compartment, this will lead to the stretch of lymphatic capillaries. When there is stretch of lymphatic capillaries, it will lead to the contraction of smooth muscle of lymphatic capillary. Certain other factors which influence movement of fluid from interstitial compartment are

Ca.Ci. - 23/14

1. Contraction of surrounding skeletal muscles
2. Movement of parts of the body
3. Pulsations of arteries adjacent to lymphatics

- the body¹⁸ (not undergoing purificatory therapy).
- When the person is encountered with *Marmopaghata* (injury to vital parts¹⁹).
- The lady having *Vishama Prasuti*²⁰ (difficult labour)
- Pratikarma Mithyopachara*²¹ (improper following of diet and regimen after treatment).

Nidana of Agantuja Shotha

बाह्यास्त्वचो दूषयिताऽभिघातः कोष्ठाशमशस्त्राग्निविपायसाद्यैः।
आगन्तुहेतुः

Ca.Ci. – 12/7

Trauma²² to the body parts with log, stone, weapon, fire, toxic substances or metals.

Samprapti of Shotha

रूपितकफान् वायुर्दुष्टो दुष्टान् बहिः सिराः।
नीत्वा रुद्धगतिस्तैर्हि कुर्यात्त्वयांससंश्रयम्॥१॥

Ma.Ni. – Shotha/1-2

उत्सेधं संहतं शोथं तमाहुर्निचयादतः।

- There will be vitiation of *Vata, Rakta, Pitta and Kapha*²³
- Vitiated *Dosha* will come to the exterior in between the *Trak* and *Mamsa* by obstruction.
- Swelling will be manifested in the body parts where obstruction of the vitiated *Dosha* taking place.

Samprapti Ghataka

- **Dosha** : *Tridosha* involvement
- **Dushya** : *Rasa, Rakta, Mamsa*
- **Srotas** : *Rasavaha, Raktavaha* and *Mamsavaha Srotas*
- **Srotodushti** : *Sanga* and *Vimargagamana*
- **Agni** : *Jatharagnimandya* in case of *Nija Shotha*
- **Ama** : *Jatharagnimandya* in case of *Nija Shotha*
- **Udbhava Sthana** : *Pakvashaya*
- **Sanchara Sthana** : *Sarva Sharira*
- **Vyakta Sthana** : *Trak*
- **Rogamarga** : *Abhyantara, Bahya* in the order of clinical manifestation
- **Svabhava** : *Ashukari* or *Chirakari*
- **Sadhyasadyata** : *Krichrasadhyata* to *Asadhyata*

Purvarupa of Shotha

तत्पूर्वरूपं दवथुः सिरायामोऽङ्गगौरवम्॥३॥

Ma.Ni. – Shotha/3

ऊष्मा तथा स्यादवथुः सिराणामायाम इत्येव च पूर्ववरूपम्।

Ca.Ci. – 12/10

- **Davathu**²⁴: burning sensation in the affected part
- **Sirayama**²⁵: visible dilatation of the blood vessels, vein in particular
- **Anga gourava**²⁶: heaviness of the body parts

4. Compression of the tissues by objects outside the body
By observing above factors influencing movement of fluid in interstitial compartment, those people who have sedentary lifestyle, we can see less movement of fluid from interstitial compartment leading to the manifestation of oedema.
- 18 When an individual is having improper administration of Panchakarma therapy, that will lead to the circulation of inflammatory markers throughout the body, one of the major symptoms observed in this particular situation could be itching all over the body leading to anaphylactic reaction. This is considered to be an allergic reaction which leads to increased capillary permeability.
- 19 Injury to any of the to vital parts will lead to traumatic shock leading to hypovolemia. Without any loss of blood volume, can lead to increased vascularity so much that even the normal amount of blood becomes incapable of filling the circulatory system adequately. This pathological mechanism can occur due to the sudden loss of vasomotor tone throughout the body leading to the massive dilatation of veins. This may lead to the appearance of oedema of the distal parts of the body.
- 20 This term refers to the appearance of the Shotha after improper diet after undergoing the treatment. Let us take the example of wound in the lower limb. It is being said that the person who is having the appearance of wound, he should avoid the substances prepared from black gram. Due to the consumption of black gram items, he is likely to have appearance of pus formation in the wound in the body.
- 21 This term refers to the appearance of the Shotha after improper diet after undergoing the treatment. Let us take the example of wound in the lower limb. It is being said that the person who is having the appearance of wound, he should avoid the substances prepared from black gram. Due to the consumption of black gram items, he is likely to have appearance of pus formation in the wound in the body.
- 22 Trauma can lead to two different types of abnormalities of cardiac output:
 1. Cardiac abnormalities which reduce cardiac output due to the compromise in the pumping ability of heart
 2. Factors which reduce venous return also reduce cardiac output due to poor venous return to the heart.
 Both the phenomena will lead to the transfer of the fluid from intravascular to interstitial compartment, thus causing the mechanism of oedema in the affected part.
- 23 In a disease, presence of pain indicates the dominance of Vata Dosha, presence of redness may imply the association of Pitta and Rakta, local raise of temperature indicates the dominance of Pitta and the appearance of swelling suggests the vitiation of Kapha Dosha during the course of the illness.
- 24 This is attributable to the excess fluid movement into the interstitial compartment from intravascular compartment. In other words, there will be increased blood supply to the affected part, leading to the raise of body temperature. Thus, the person may have burning sensation of the affected part.
- 25 Sirayama is attributable to the increased blood supply along with development of collaterals of the blood vessels. This will lead to the redness of the area afflicted.
- 26 Anga Gourava is attributable to the transfer of fluid from intravascular to interstitial compartment and excess of proteins in the interstitial compartment. Thus, the person may have heaviness of the affected parts. Due to the derangement of Sterling forces, more amount of fluid

Samanya Lakshana of Shotha

समीरचं स्यादनपस्थितत्वं सोत्सेधमध्माऽथ सिरातनुत्वम्।
सलोमहर्षश्च विषर्णता च सामान्यलिङ्गं श्वयथोः प्रदिष्टम्॥

Ma.Ni. – Shotha/6; Ca.Ci. – 12/11

- ✱ **Sagourava**²⁷: heaviness of the body parts
- ✱ **Anavasthitatva**²⁸: unsteadiness of the swelling
- ✱ **Utsedha**²⁹: globular swelling
- ✱ **Ushma**³⁰: local or generalized body temperature
- ✱ **Siratanutva**: thinning of the blood vessels
- ✱ **Lomaharsha**: horripulation
- ✱ **Vivarnata**³¹: discolouration

Bheda of Shotha

सर्वे हेतु विशेषस्तु रूपभेदान्नवात्मकम्॥2॥

दोषैः पृथग्द्रव्यैः सर्वैरभिघाताद्विषादपि।

Ma.Ni. – Shotha/2-3

त्रिविधो निजश्च सर्वाधगात्रावयवाधितत्वात्॥7॥

Ca.Ci. – 12/7

Types of Shotha

9 types based on *Dosha* dominance and other etiological factors.

1. *Vataja*
2. *Pittaja*
3. *Kaphaja*
4. *Vatapittaja*
5. *Pittakaphaja*
6. *Vatakaphaja*
7. *Sannipataja*
8. *Abhigataja*
9. *Vishaja*

Based on the appearance on the body part of the patients,
3 types of *Nija Shotha*.

1. *Sarva*³² (swelling all over the body)
2. *Ardhagatraja*³³ (swelling in half side of the body)
3. *Avayava*³⁴ (swelling in one part of the body)

Vataja Shotha

चलस्तनुत्वकपरुषोऽरुणोऽसितः प्रसुप्तिहर्षातिर्युतोऽनिमित्ततः।

प्रशाम्यति प्रोन्नमति प्रपीडितो दिवाली च श्वयथुः समीरणात्॥12॥

Ca.Ci. – 12/120

- ✱ **Chala**: unstable swelling
- ✱ **Tanu Tvak**: thinning of the skin
- ✱ Rough skin
- ✱ Reddish or blackish discolouration of the swelling
- ✱ Associated with numbness, horripulation, severe pain without any identifiable reason
- ✱ Rapid disappearance of the pitting on pressure when we do pitting on pressure test
- ✱ Worsening of the symptoms in day time

Pittaja Shotha

मृदुः सगन्धोऽसितपीतरागवान् भ्रमज्वरस्वेदतृषामदान्वितः।

य उप्यते स्पर्शरुग्झिरागकृत् स पित्तशोथो भृशदाहपाकवान्॥13॥

Ca.Ci. – 12/13

- ✱ Soft swelling
- ✱ Foul smell from the swelling
- ✱ Blackish, yellowish discolouration and redness of the swelling
- ✱ Giddiness
- ✱ Fever
- ✱ Excess sweating
- ✱ Thirst
- ✱ Altered state of consciousness

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- id move from the capillaries to the interstitial spaces, there will sluggish movement of the blood in the small vessels happens. This is called sludging of the blood. Because of this phenomenon, the person will have heaviness of the body parts.
- 27 As per Ayurveda fundamentals, Rupa (clinical feature) is the complete manifestation of Purvarupa (prodromal symptom). Gouravata is the complete manifestation of prodromal symptom. This is attributable to the excess accumulation of fluid in the interstitial compartment as well as sluggish movement of small blood vessels (sludging of blood). Both of the above pathological events resulting in the heaviness of the body parts.
- 28 To understand this concept, we have to understand the concept of Sterling forces. Abnormal hydrostatic pressure and oncotic pressure in the capillaries and interstitial spaces are responsible for the increased accumulation of fluid in the interstitial spaces. Let us learn the concept with an example. If the concentration of protein is increased in the interstitial spaces of more than 4 mmHg, that will lead to more accumulation of fluid in the interstitial space and reduction of the protein in the interstitial space will lead to the reduction of fluid in the interstitial spaces. Thus, depending upon the concentration of the protein, salts in the interstitial compartment and variations in hydrostatic pressure may lead to variations in the swelling of the affected parts.
- 29 This term is attributable to the increase of interstitial space of the affected part. This occurs due to the excess movement of fluid from intravascular to interstitial compartment.
- 30 This term is attributable to increased vascularity to the affected part where Shotha is present.
- 31 The term Vivarnata refers to discolouration. The discolouration can be either redness or bluish or yellowish discolouration or pallor.
- 32 As per current understanding, generalised oedema is termed as anasarca or dropsy. This is seen in the patients having cardiac oedema, renal oedema, hepatic oedema and hypoproteinemia.
- 33 Swelling in half side of the body may be observed in the patients suffering from hemiplegia or poor usage of the half of the body.
- 34 This is observed in case of localised disease manifestation.

- Burning sensation
- Severe pain on touching the site of the swelling
- Redness of the eyes
- Rapid development of pus formation and rapid appearance of burning sensation of the site of the swelling

Kaphaja Shotha

गुरुः स्थिरः पाण्डुरोचकान्वितः प्रसेकनिद्रावमिवह्लिमान्द्रुकृत्।
सकृच्छ्रजन्मप्रशमो निपीडितो न चोमेद्रात्रिबला कफात्मकः॥14॥

Ca.Ci. - 12/14

- Heaviness of the swollen part of the body
- Firm swelling of the affected part
- Yellowish white discolouration
- Tastelessness
- Excessive salivation
- Excess sleep
- Vomiting
- Reduced digestive fire
- Slow pitting type of oedema i.e. slow disappearance of pitting of the oedematous part when we perform the test of pitting on pressure.
- Worsening of the symptoms during night time

Clinical manifestation and site of the lesion as per Madhava Nidana

दोषाः श्रयथुमूर्ध्वं हि कुर्वन्त्यामाशयस्थिताः॥15॥

पकाशयस्था मध्ये तु वर्चः स्थानगतास्त्वधः।

कृत्स्नदेहमनुप्राप्ताः कुर्युः सर्वसरं तथा॥16॥ Ma.Ni. - Shotha/15-16

- Dosha situated in the Amashaya responsible for the manifestation of Shotha in the upper part of the body.
- Dosha present in the area between Amashaya and Pakvashaya lead to the occurrence of Shotha in middle part of the body.
- Dosha located in the Pakvashaya will lead to the appearance of Shotha in the lower part of the body.

Sukhasadhya Lakshana of Shotha

- 35 This term refers to the presence of pleural or pericardial effusion or large ascites. When a clinician observes the swelling in the abdomen, it indicates either hepatic oedema. This may occur as a result of either alcoholic liver disease or any intraabdominal malignancy.
- 36 Presence of swelling all over the body is termed as anasarca or dropsy. This is seen in the patients of serious cardiac compromise (cardiac oedema), chronic renal failure (renal oedema), hypoproteinemia. It can also be considered as Sannipataja Shotha.
- 37 Swelling in half of the body observed in the persons who are suffering from Pakshaghata or cerebro vascular accident. This is considered to be due to poor mobility of the limbs.
- 38 यस्तु पादाभिनिर्वृतः शोथः सर्वाङ्गो भवेत्। जन्तोः स च सुकष्टः स्यात् प्रसृतः स्त्रीमुखाश्च यः 16
- 39 यद्यपि गुहाप्रभयः स्त्रिया या पुरुषस्य वा। सच कतमो ज्ञेयो यख च स्युरुपद्रयाः॥17॥
- 40 When you soeak about the appearance of swelling from feet to upper part of the body, the parts of the body to be examined are feet,

The following Lakshana are considered as considered to be having good prognosis in Shotha. They are

अहीनमांसस्य य एकदोषजो नवो बलस्थस्य सुखः स साधने।

Ca.Ci. - 12/16

- When the person is not having Mamsa Kshaya
- When one Dosha is involved in the manifestation of Shotha
- If the duration of illness is short or disease is of recent origin
- If the Rogibala (strength of the patient) is more than Rogabala (strength of disease)

Asadhya Lakshana of Shoth

The following Lakshana are considered to be having poor prognosis as far as Shotha is concerned. They are

यो मध्यदेशे श्रयथुः ख कष्टः सर्वगस्य यः।
अर्धाङ्गे रिष्टभूतः स्यायश्चोर्ध्वं परिसर्पति॥17॥
श्वासः पिपासा उर्दिध दीर्घत्वं ज्वर एव च।
यस्य चात्रे रुचिर्नास्ति श्रयथुं तं विवर्जयेत्॥18॥
अनन्योपद्रवकृतः शोथः पादसमुत्थितः।
पुरुष हन्ति नारी च मुखजो गुजो द्वयम्।
नवोऽनुपद्रथः शोथः साध्योऽसाध्यः पुरेरितः॥19॥
विवर्जयेत्कुशयुद्राश्रितं च तथा गले मर्मणि संधितं च।
स्थूलः सरब्धापि भवेद्वियज्योयचापि बालस्थविराबलानां॥20॥

Ma.Ni. - Shotha/17-20

कृशस्य रोगैरवस्य यो भवे परित्वेदीनवलस्य सर्वगः॥15॥

Ca. Ci. - 12/15

कृशस्य रोगैरवलस्य यो भवेदुपद्रवैर्वा वमिपूर्वकैर्युतः।

सहन्ति मर्मानुगतोऽथ राजिमान् परित्वेदीनवलस्य सर्वगः॥15॥

Ca.Ci. - 12/15

- If the Shotha is present in the middle part of the body³⁵ (thorax and abdomen)
- If Shotha is present all over the body³⁶
- If half of the body³⁷ is having Shotha is observed in the patient
- In males³⁸, the swelling spreading from feet to upper part³⁹ of the body and⁴⁰

Ca.Su. - 18/16,17

- ❖ In females, if the swelling spreads from face to the lower part⁴¹ of the body.
- ❖ In males and females, if the swelling starts from genitals⁴², then such type of Shotha is having poor prognosis.
- ❖ If it is associated with breathing discomfort⁴³, thirst⁴⁴, vomiting, weakness, raised body temperature, loss of taste.
- ❖ If Shotha is associated with other Upadrava⁴⁵ (complication)
- ❖ If Shotha is associated with other Upadrava⁴⁶ (complication)
- ❖ If the Shotha is involving abdomen⁴⁷, throat⁴⁸ and vital points (Marma)⁴⁹, if the Shotha is large and having rough⁵⁰ appearance
- ❖ If the swelling⁵¹ is present in children, elderly people and weak individuals

Key points of the chapter

सर्वमुत्सेधं शोफमित्याहु Atanka Darpana, Ma.Ni. – 36/1,2. As per the commentary Atanka Darpana, any kind of the swelling manifested in any part of the body is termed as Shotha or Shophra.

- ❖ Shotha occurs in between Tvak and Mamsa Dhatu.
- ❖ It can occur in one Avayava (localised) or Dehardha (half of the body) or Sarvasharira (generalized).
- ❖ As far as Sankhya Samprapti of Shotha is concerned, Shotha is of 9 types:
- ❖ Vataja, Pittaja, Kaphaja, Vata-pittaja, Vata-kaphaja, Pitta-kaphaja, Sannipataja, Abhigataja and Vishaja.
- ❖ Divabali (worsening of the symptoms in day time) is the specific feature of Vataja Shotha and Ratri-bali (worsening of the symptoms in night time) is the specific feature of Kaphaja Shotha.
- ❖ As far as the manifestation of Shotha is concerned, there will be excess accumulation of fluid in one part of the body or whole body.
- ❖ This is especially seen in the individual who are suffering from expanded interstitial compartment.

abdomen, chest, orbital region. Analysis of cardiac dysfunction can be done on assessment of oedema of feet; swelling in the abdominal region suggests the presence of hepatobiliary dysfunction; presence of pulmonary oedema implies the compromise of respiratory as well as cardiac dysfunction; oedema in the periorbital region implies abnormal renal function. In females, the analysis should be done in the similar fashion.

- 41 This is attributable to severe progression of the disease. This type of clinical presentation can be seen in the chronic renal failure. Usually the person with renal failure will be presented with periorbital oedema in early morning time and later he will be presented with generalised oedema due to hypoproteinemia.
- 42 When the person is having swelling arising from genitals, the earlier clinical manifestation can be observed in absence of micturition i.e. anuria. If urine output of one kidney is diminished, the other kidney is taking charge of the function of affected kidney. If there is urethral obstruction or bilateral ureteric obstruction, that will be leading to the manifestation of hydronephrosis and hydronephrosis. This will lead to the manifestation of uremia. This results in the manifestation of oedema secondary to sodium and water retention.
- 43 Breathing discomfort can be seen in conditions where there is excess accumulation of fluid in peritoneal cavity which leads to compression of diaphragm upwards or there is accumulation of fluid in alveoli which leads to the compromise in respiratory function or inflammatory reaction happening in either upper or lower respiratory passages where we can see inflammatory oedema in the affected part.
- 44 The presence of thirst indicates condensed intracellular or interstitial or intravascular compartment. This is observed as a sequel of any of the infection occurring in any part of the body. As far as Shotha is concerned, there will be redistribution of the fluid in oedematous part leading to the accumulation of fluid in affected part and shrunken interstitial compartment in other parts. This leads to the manifestation of thirst commonly.
- 45 छर्दिः श्वासोऽरुचिस्तृष्णा ज्वरोऽतीसार एव च। सप्तकोऽयं सदैर्बल्यः शोफोपद्रवसंग्रहः॥१८॥
Vomiting, breathlessness, tastelessness, thirst, fever, diarrhoea, weakness are Upadrava of Shotha as per Charaka Ca.Su. – 18/18
- 46 छर्दिः श्वासोऽरुचिस्तृष्णा ज्वरोऽतीसार एव च। सप्तकोऽयं सदैर्बल्यः शोफोपद्रवसंग्रहः॥१८॥
Vomiting, breathlessness, tastelessness, thirst, fever, diarrhoea, weakness are Upadrava of Shotha as per Charaka Ca.Su. – 18/18
- 47 The generalised swelling observed in the abdominal region is termed as ascites. As per western medicine, this is nothing but oedema in the serous cavity.
- 48 Acute angioedema is usually developed in this particular region. This is anaphylaxis reaction which is usually seen in the patients of urticaria. This is characterised by locally diffuse and painful soft tissue swelling which is usually asymmetric. In the case of involvement of tongue etc, the person will have respiratory distress and stridor which should be treated as emergency.
- 49 To understand this concept, refer the pathophysiology observed in case of Agantuja Shotha.
- 50 Rough appearance of the swelling indicates the chronicity of the swelling or Shotha. For example, in case of elephantiasis, we can observe the rough skin like skin of an elephant.
- 51 The swelling in these age groups can be observed in protein malabsorption either in children, elders and weak individuals. In this situation, there will be excess presence of fluid in interstitial compartment than intravascular compartment. The nature of oedema is slow pitting type of oedema i.e. disappearance of oedema can occur after 60 seconds.

Long essays

- Explain Nidana Panchaka of Shotha
- Describe Nidana, Purvarupa, Samanya Lakshana, Vidhi Samprapti, Asadhya Lakshana and Sadhyasadyata of Shotha

Short essays

- Explain the Lakshana of Vataja Shotha
- Explain the Lakshana of Kaphaja Shotha
- Explain in brief about the types of Shotha.

MCQs

1. Ratrikali is the specific feature of Shotha.

- a. Vataja; b. Kaphaja;
- c. Pittaja; d. Sannipataja

Answer: (b)

2. Sankhya Samprapti of Shotha as per Madhava Nidana is

- a. 4 types; b. 7 types;
- c. 9 types; d. 12 types

Answer: (c)

3. Davathu or burning sensation in the affected part is considered asof Shotha.

- a. Upadrava; b. Purvarupa;
- c. Nidana; d. Samprapti

Answer: (b)

4. Sarva, Ardhang and Avayava are the types of

- a. Raktapitta; b. Amlapitta;
- c. Shitapitta; d. Shotha

Answer: (d)

5. Abhighataja is the type of

- a. Shotha; b. Apasmara;
- c. Grahani; d. Kasa

Answer: (a)

6. The earliest sign observed in the patient of alcoholic liver disease is

- a. Bilateral pedal oedema;
- b. Periorbital oedema;
- c. Ascites;
- d. Cough with pink frothy sputum

Answer: (c)

7. The presence of pitting type of oedema in both feet indicates

- a. Renal oedema;
- b. Hepatic oedema;
- c. Brawny oedema;
- d. Cardiac oedema

Answer: (d)

8. The approximate number of capillaries in adult human body is

- a. 10 million;
- b. 10 billion;
- c. 10 trillion;
- d. 100,000 capillaries

Answer: (b)

9. Cough with pink frothy sputum indicates the dysfunction of

- a. Kidney;
- b. Liver;
- c. Heart;
- d. Hypoproteinemia

Answer: (c)

10. The term Divabali refers to

- a. Aggravation of the symptoms in day time;
- b. Relief of the symptoms in day time;
- c. Relief of the symptoms in night time;
- d. Aggravation of the symptoms in night time.

Answer: (a)

Chapter resources

- <https://yamunapharmacy.com/edema-or-oedema>
- Madhava Nidana – 36th chapter Shotha Nidana
- Charaka Samhita Sutra Sthana 18th chapter – Trishothiya Adhyaya
- Charaka Samhita Chikitsa Sthana 12th chapter – Shotha Chikitsa Adhyaya
- Sushruta Samhita, Chikitsa Sthana 23rd chapter
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006